Retrieval-Augmented Question Answering on the 1988 French Political Manifestos

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Abstract

This project builds a Retrieval-Augmented Generation system for question answering over the 1988 subset of the Archelec corpus, a collection of French political manifestos from the Fifth Republic. The objective is to allow users to ask natural language questions about the 1988 manifestos and obtain answers grounded in the retrieved documents.

The pipeline combines corpus loading, exploratory data analysis, text cleaning, chunking, vector embeddings, FAISS-based semantic search, and an instruction-tuned large language model. The retrieval step identifies relevant passages from the corpus, while the generation step produces a short factual answer based on these passages. The language model used for generation is `meta-llama/Llama-3.1-8B-Instruct`, accessed through a Hugging Face endpoint.

Although text files were available for several years, the project focuses on 1988 because of computational constraints on SSPCloud, especially the cost of computing embeddings and building the FAISS vector database. The final corpus contains 3,628 text files and is split into 16,815 chunks. A small evaluation comparing RAG with a LLM-only baseline shows that the RAG answers are more grounded in the retrieved corpus, with an average groundedness score of 0.585 compared with 0.148 for the LLM-only baseline.

1 Introduction

Political manifestos constitute a central source for the study of political discourse. They allow candidates to present their priorities, justify their positions, and communicate directly with voters. In the context of French legislative elections, these documents are particularly valuable because they combine national political narratives with local campaign dynamics.

The Archelec corpus provides a large-scale digitized collection of French electoral manifestos from the Fifth Republic. However, this type of corpus is difficult to explore manually. It contains thousands of documents, with substantial heterogeneity in length, style, and structure. In addition, the texts are derived from OCR transcriptions, which may introduce noise at the word or sentence level. As a result, identifying relevant information on a specific topic such as unemployment, education, Europe, or immigration requires substantial effort.

Beyond this difficulty, the corpus also presents structural complexity. Documents differ significantly across electoral rounds, geographical areas, and levels of detail. Some manifestos contain long and structured policy discussions, while others are shorter and more focused on campaign messaging. These differences make simple keyword-based exploration inefficient and motivate the use of more advanced Natural Language Processing techniques.

This project addresses the following question:

How can we build a system that answers natural language questions about the 1988 political manifestos using evidence retrieved from the corpus?

To answer this question, we implement a Retrieval-Augmented Generation (RAG) pipeline. The system first transforms the corpus into a vector database by splitting documents into smaller text chunks and embedding them into a semantic space. Given a user query, the system retrieves the most relevant chunks using similarity search. These retrieved passages are then provided to a large language model, which generates an answer grounded in the retrieved context.

Before building the RAG system, we perform an extensive exploratory analysis of the corpus. This analysis highlights key characteristics of the data, including heterogeneity in document length, differences between electoral rounds, lexical diversity, and the overall quality of OCR transcriptions. These elements are essential to justify the design choices of the pipeline, particularly the use of chunking and semantic retrieval.

Finally, we propose a simple quantitative evaluation of the system by comparing the RAG approach to a baseline where the language model answers questions without access to retrieved context. The results show that the RAG pipeline produces answers that are more grounded in the corpus and better aligned with the retrieved evidence.

The objective of this project is therefore not to produce a complete political or historical analysis of the 1988 elections. Instead, it aims to build a practical and interpretable question-answering system that allows users to explore the corpus efficiently while preserving a strong link between generated answers and original textual sources.

2 Corpus and Data

2.1 The Archelec Corpus

The Archelec corpus is a collection of French electoral manifestos from the Fifth Republic. These documents were collected and digitized by Sciences Po and made available through the Archelec project. In the NLP course, text files were provided for several election years, including 1973, 1978, 1981, 1988, and 1993.

In this project, we use only the 1988 subset.

2.2 Choice of the 1988 Subset

The restriction to 1988 is mainly due to computational constraints. The project was implemented on SSPCloud, where only limited GPU resources were available. Building a RAG system requires computing embeddings for a large number of text chunks before creating the FAISS vector database. Applying this process to all available years would have been too slow in the available environment.

Using only the 1988 corpus made it possible to build and evaluate a complete pipeline while keeping the computation manageable. It also gives the project a clear scope: the system answers questions about one electoral year rather than mixing several political contexts.

2.3 Data Format

The input data consists of raw text files. Each file corresponds to one manifesto or one transcribed electoral document. The notebook loads all `.txt` files from the directory `text_files`. Empty files are ignored.

The final corpus contains:

Metric	Value
Number of documents	3,628
Total number of words	2,002,003
Average number of words per document	551.82
Median number of words per document	495
Minimum number of words	9
Maximum number of words	2,552

Table 1: Basic statistics of the 1988 corpus

For each document, the notebook stores the file name, the raw text, the number of characters, the number of words, and the number of lines.

3 Exploratory Data Analysis

Before building the RAG pipeline, we performed a descriptive analysis of the corpus. This step is important because retrieval quality depends on the structure, size, vocabulary, and noise level of the underlying documents. Since the goal of the project is to answer questions from the 1988 political manifestos, the exploratory analysis also helps justify the main technical choices of the pipeline, especially chunking and semantic retrieval.

3.1 Corpus Structure and Electoral Rounds

The first step was to parse the file names in order to recover useful metadata. Two types of documents coexist in the raw corpus: *professions de foi* and ballot-related files. Since the objective of this project is to analyze political manifestos, we retained the *professions de foi* files for the metadata analysis.

The resulting subset contains 3,544 *profession de foi* files. These documents cover 101 departments and 566 constituencies, which indicates that the corpus is geographically broad and not restricted to a small number of local electoral contexts.

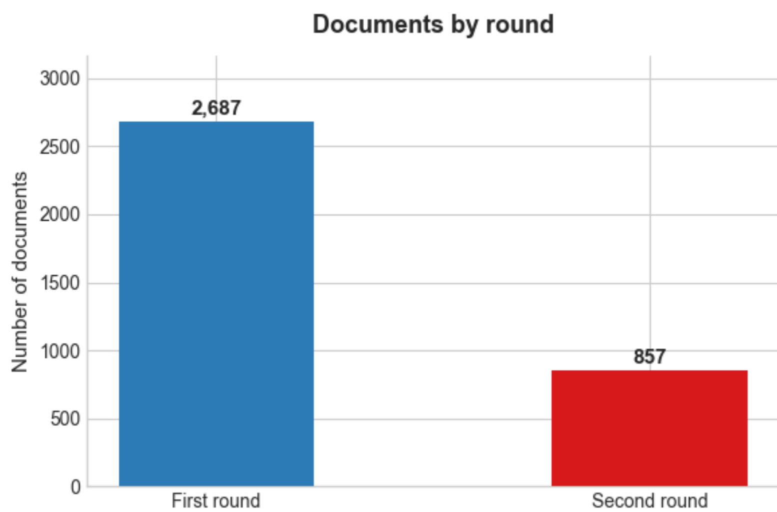

Figure 1: Distribution of documents by electoral round. First-round documents represent most of the corpus.

Figure 1 shows a clear imbalance between electoral rounds. The first round contains 2,687 documents, corresponding to 75.8% of the *profession de foi* subset, while the second round contains 857 documents, corresponding to 24.2%. This imbalance is important because global descriptive statistics are mainly driven by first-round documents. It also suggests that retrieval results may be more likely to come from first-round material simply because this part of the corpus is larger.

3.2 Geographical Coverage

The corpus provides broad territorial coverage. However, the number of available documents differs across departments. This variation may come from differences in the number of constituencies, the number of candidates, or the availability and digitization of archival documents.

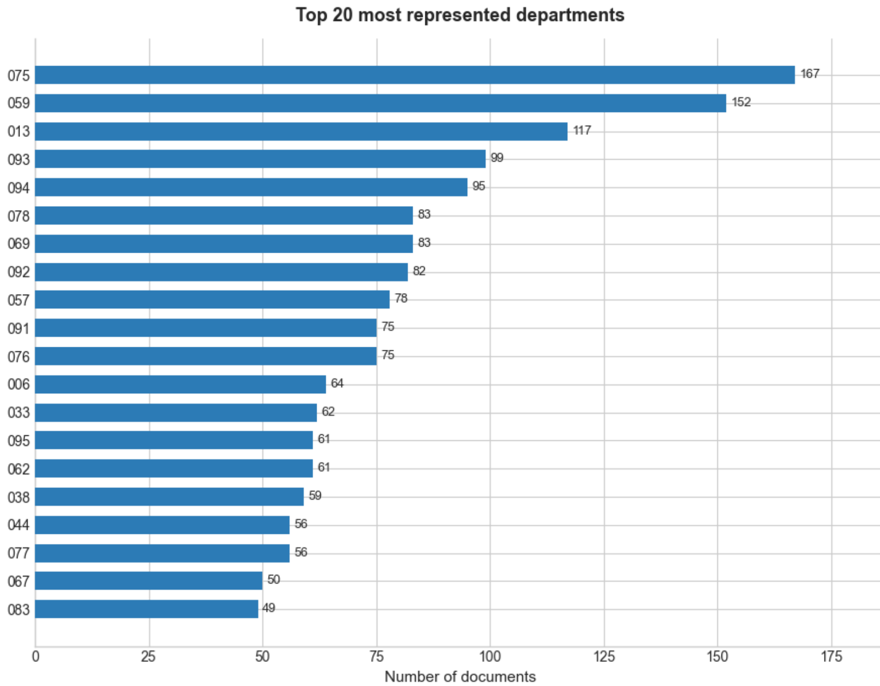

Figure 2: Top 20 most represented departments in the 1988 corpus.

Figure 2 shows that some departments are more represented than others. This does not prevent the use of the corpus for question answering, but it means that the RAG system should be interpreted as a tool for exploring the available archive, not as a perfectly balanced statistical sample of all French political discourse in 1988.

3.3 Document Length

Document length is a key issue for retrieval. If complete documents are used directly as retrieval units, long manifestos may contain several unrelated ideas in the same vector representation. On the other hand, if text units are too small, they may lose important context. For this reason, we first analyzed the distribution of document lengths before deciding on the chunking strategy.

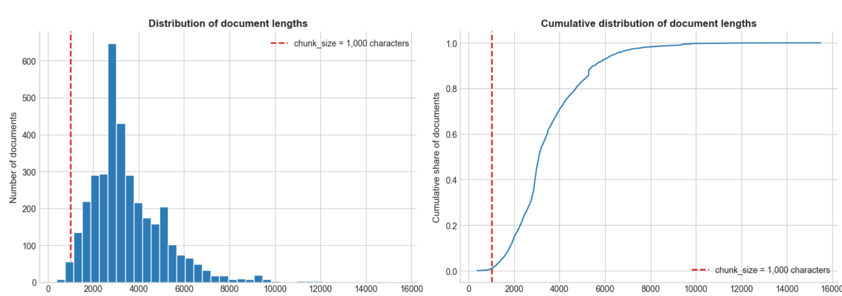

Figure 3: Distribution of document lengths in the 1988 corpus.

Figure 3 shows that most documents are concentrated around a few hundred words, but the distribution is not homogeneous. Some manifestos are much longer and contain more developed political argumentation. This supports the use of chunking before retrieval: instead of embedding entire documents, the system embeds smaller passages that are more likely to focus on a specific topic.

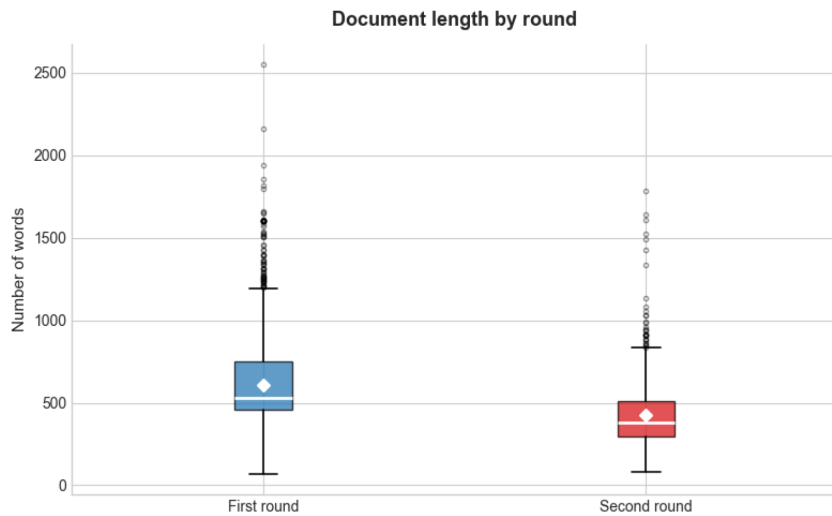

Figure 4: Document length by electoral round.

Figure 4 compares document length by electoral round. First-round documents are longer on average, with 606.4 words against 423.0 words for second-round documents. The median length is also higher in the first round, with 527 words against 381 words in the second round. A Mann-Whitney test gives a p-value close to zero, indicating that the difference in document length between the two rounds is statistically significant.

This result is substantively meaningful. First-round manifestos tend to be more programmatic and detailed, while second-round documents may be more strategic, shorter, and focused on mobilization. For retrieval, this matters because first-round documents are likely to provide richer context for broad policy questions, whereas second-round documents may be more concise and campaign-oriented.

3.4 Text Cleaning for Exploration

For the exploratory analysis, we used a simple cleaning function. The preprocessing included lowercasing, removing punctuation and digits, tokenization, stopword filtering, and keeping words with more than two characters.

This cleaning was used only for descriptive statistics, word frequencies, lexical diversity, and the automatic evaluation metric. It was not used as a replacement for the original text in the RAG pipeline, because the question-answering system should preserve as much linguistic information as possible. In particular, political expressions, punctuation, and local context may still be useful for generation.

3.5 Text Quality Indicators

Because the corpus comes from digitized archival documents, it is necessary to check whether the texts contain strong OCR noise. We computed three simple indicators: the share of alphabetic characters, the share of digit characters, and the share of very short lines.

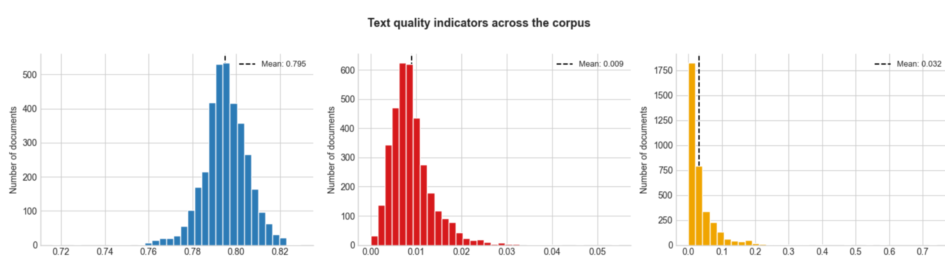

Figure 5: Text quality indicators across the corpus.

Figure 5 suggests that the corpus is generally usable for NLP. The mean alphabetic character ratio is 0.7951, while the mean digit ratio is only 0.0090 and the mean short-line ratio is 0.0317. These values indicate that the corpus mostly contains regular textual content rather than numerical or heavily fragmented material.

This does not mean that the OCR is perfect. Some word-level transcription errors may still exist and can affect embeddings or generation. However, the indicators do not suggest a massive OCR quality problem at the corpus level.

3.6 Most Frequent Words

After cleaning, we computed the most frequent words in the corpus. This gives a first view of the dominant vocabulary used in the 1988 manifestos.

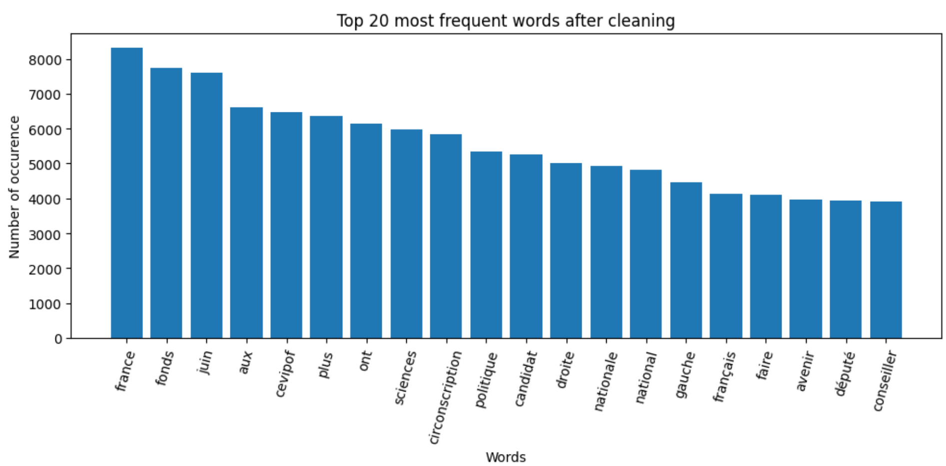

Figure 6: Top 20 most frequent words after cleaning.

Figure 6 shows that the vocabulary is strongly political and institutional. Frequent terms include words related to France, politics, candidates, ideological camps, voters, and public life. This confirms that the corpus contains the expected type of electoral discourse.

However, word frequency remains a limited analysis. It shows what words appear often, but it does not tell us how a topic is discussed or whether a passage answers a specific question. For example, knowing that a word such as *France* is frequent does not explain whether it is used in a national, economic, institutional, or identity-related argument. This limitation motivates the use of semantic retrieval.

3.7 Most Frequent Bigrams

To go beyond isolated words, we also computed the most frequent bigrams. Bigrams capture recurring two-word expressions and therefore provide a slightly richer view of the corpus.

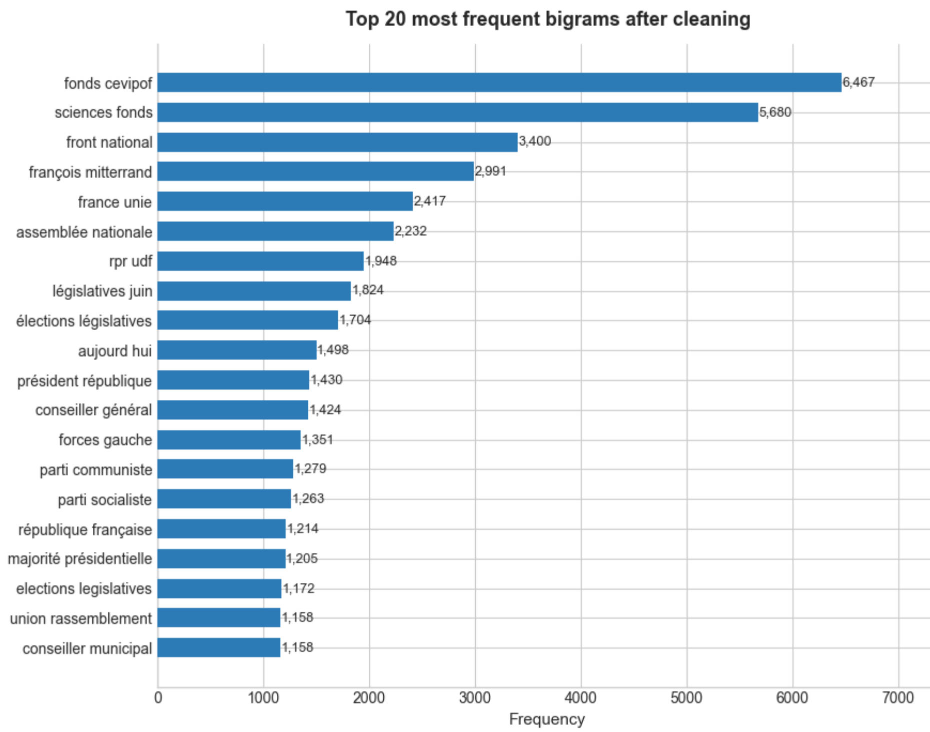

Figure 7: Top 20 most frequent bigrams after cleaning.

Figure 7 highlights recurrent expressions in the manifestos. These expressions are useful because political discourse is often built around repeated formulas, institutional phrases, and campaign slogans. Nevertheless, bigrams remain a surface-level tool. They cannot capture longer semantic dependencies or answer open-ended questions, which again supports the use of a RAG framework.

3.8 Distinctive Terms by Electoral Round

We also compared the most distinctive terms in first-round and second-round documents using TF-IDF scores. The objective was to identify whether the vocabulary changes between rounds.

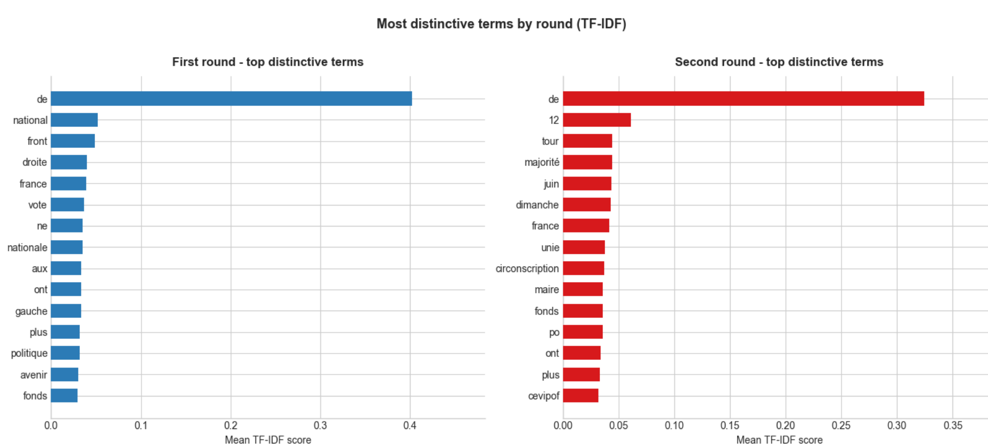

Figure 8: Most distinctive terms by electoral round using TF-IDF.

Figure 8 shows that first-round and second-round documents do not emphasize exactly the same vocabulary. First-round documents tend to be more programmatic, with a broader range of policy-related terms. Second-round documents are more concentrated around mobilization and electoral competition.

This difference matters for the RAG system. A query about a policy theme may retrieve longer first-round passages, while a query related to electoral confrontation or voting strategy may retrieve more second-round material. The retrieval system therefore reflects not only semantic similarity but also the structure of the campaign itself.

3.9 Lexical Diversity

We computed lexical diversity at two levels. First, at the corpus level, lexical diversity is defined as:

$$\text{Lexical Diversity} = \frac{\text{Number of unique cleaned words}}{\text{Total number of cleaned tokens}}.$$

The corpus contains 44,963 unique cleaned words and 1,075,337 cleaned tokens, giving a corpus-level lexical diversity of 0.0418. This value is expected for a large corpus: many political and institutional words are repeated across documents, while a large number of less frequent words appear only occasionally.

We also computed the Type-Token Ratio (TTR) at the document level. For a document d, it is defined as:

$$\text{TTR}(d) = \frac{\text{Number of unique cleaned words in } d}{\text{Number of cleaned tokens in } d}.$$

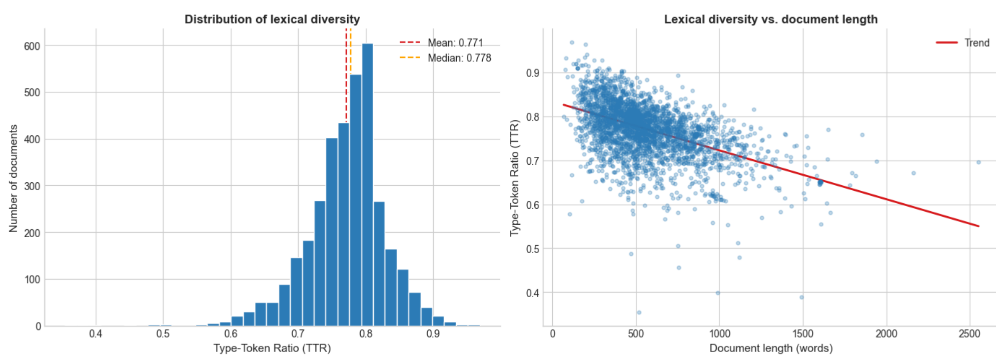

Figure 9: Distribution of document-level lexical diversity and relationship between TTR and document length.

Figure 9 shows that the average TTR is 0.7713 and the median TTR is 0.7778. These values suggest that many documents use a varied vocabulary. The scatter plot also shows the expected negative relationship between document length and TTR: longer documents tend to have lower TTR because repeated words become more likely as text length increases.

This point is important for retrieval. High lexical diversity provides useful signals for embeddings, because documents contain enough variation to be semantically distinguished. However, TTR must be interpreted carefully because shorter documents mechanically tend to have higher values.

3.10 Zipf Distribution

We also analyzed the word frequency distribution using Zipf's law. In natural language, the frequency of a word is usually inversely related to its rank: a few words occur very often, while most words are rare.

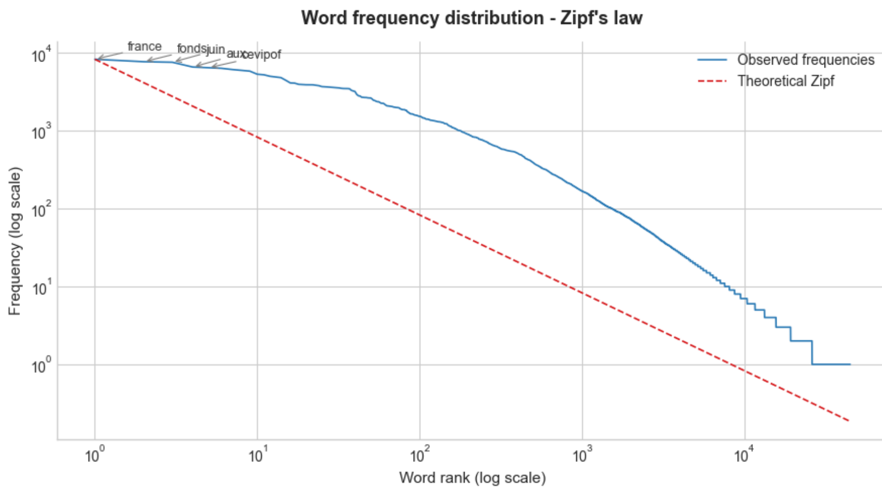

Figure 10: Word frequency distribution compared with a theoretical Zipf reference.

Figure 10 confirms that the corpus follows the usual heavy-tailed structure of natural language. This is reassuring because it suggests that the corpus behaves like a standard textual dataset despite being produced from OCR transcriptions of electoral documents.

For the RAG system, the long tail of rare words is also useful. Rare terms may correspond to local issues, names, institutions, or specific policy topics. Semantic embeddings help exploit this richer vocabulary better than a simple keyword search.

3.11 TF-IDF Cosine Similarity

As an additional exploratory step, we computed pairwise cosine similarities between documents using TF-IDF representations. This provides a simple measure of lexical proximity between manifestos.

Metric	**Value**
Mean pairwise cosine similarity	0.2606
Median pairwise cosine similarity	0.2519
Maximum pairwise cosine similarity	1.0000

Table 2: Pairwise TF-IDF cosine similarity between documents

Table 2 shows moderate similarity values. The mean similarity is 0.2606 and the median is 0.2519. This indicates that the documents share a common political vocabulary but are not identical. This is exactly the kind of setting where retrieval is useful: the system must identify relevant passages for each query rather than assume that all manifestos say similar things.

We also computed a similarity heatmap on a stratified sample of 120 documents, with 60 first-round documents and 60 second-round documents.

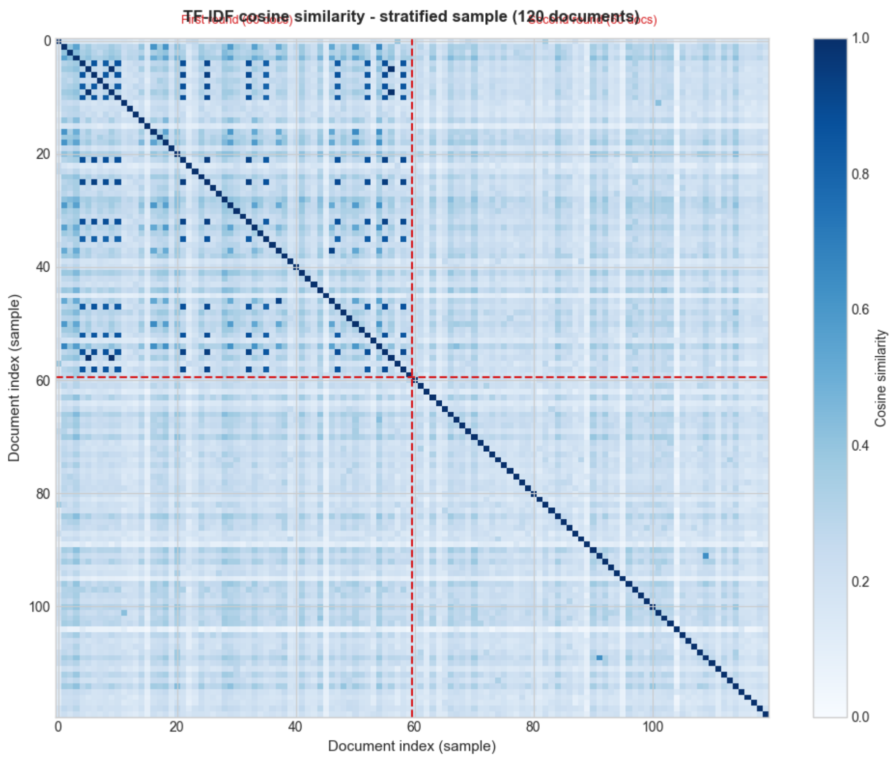

Figure 11: TF-IDF cosine similarity heatmap on a stratified sample of first- and second-round documents.

Figure 11 shows that documents from the first round are more similar to each other than documents from the second round. The mean similarity within first-round documents is 0.2803, compared with 0.2021 within second-round documents and 0.2190 between the two rounds.

This suggests that first-round manifestos share a more standardized political vocabulary, while second-round documents are more heterogeneous. This may reflect differences in campaign strategy: first-round manifestos often present broad programs, whereas second-round documents may be more local, strategic, or candidate-specific.

3.12 Chunking Analysis

We simulated the chunking step before building the final vector index.

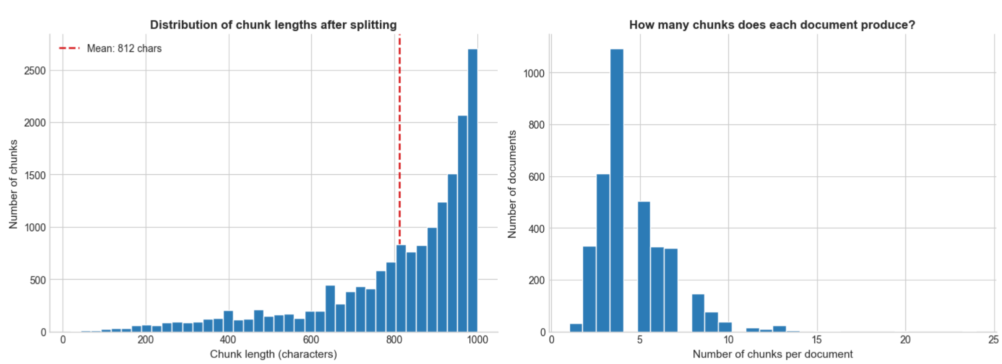

Figure 12: Distribution of chunk lengths after applying the recursive character splitter.

Figure 12 shows that the chunking strategy produces 16,688 chunks in the preview, with an average chunk length of 812.3 characters and a median length of 882 characters. The average number of chunks per document is 4.71. The final RAG index contains 16,815 chunks.

This validates the chunking strategy. The chunks are generally close to the target size, while the overlap of 200 characters helps preserve continuity between adjacent passages. This is important because political arguments can span several sentences, and cutting documents too sharply could reduce retrieval quality.

4 Related Work

4.1 NLP for Political Texts

Natural Language Processing is commonly used to analyze political texts. Typical applications include party classification, ideology detection, topic modeling, sentiment analysis, and analysis of electoral manifestos.

Traditional methods such as bag-of-words, TF-IDF, and topic modeling are useful because they are simple and interpretable. They can identify important words, compare documents, and detect broad themes. However, these methods are often based on lexical overlap and may fail to retrieve passages that express similar ideas with different words.

For a question-answering task, these limitations matter. A user does not only want a list of frequent words or similar documents. The user wants a direct answer based on the relevant parts of the corpus.

4.2 Question Answering on Documents

Question answering systems aim to produce direct answers to natural language questions. In a closed-domain setting, the answer must be grounded in a specific corpus. This is the case in this project: the model should answer questions about the 1988 manifestos, not about French politics in general.

A good system must therefore satisfy two conditions. First, it must retrieve relevant passages from the corpus. Second, it must generate an answer that remains faithful to these passages.

4.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation combines information retrieval and text generation. Instead of asking a language model to answer only from its internal knowledge, the system first retrieves relevant documents and then gives them to the model as context.

This approach is well suited to the Archelec corpus. It reduces hallucinations, allows flexible exploration, and makes the generated answer more connected to the actual documents.

5 Methodology

5.1 General Pipeline

The implemented pipeline follows several steps:

1. load the 1988 text files;
2. build a dataframe for corpus analysis;
3. clean and tokenize texts for exploratory statistics;
4. wrap each document into a LangChain `Document`;
5. split documents into chunks;
6. compute embeddings for each chunk;
7. store embeddings in a FAISS vector database;

8. retrieve relevant chunks for a user question;
9. generate an answer with a LLM using the retrieved context.

5.2 Chunking

The raw documents are split using a recursive character text splitter. The parameters are:

- chunk size: 1000 characters;
- chunk overlap: 200 characters.

After chunking, the corpus contains 16,815 chunks. The overlap helps preserve continuity between consecutive chunks, while the chunk size keeps each retrieval unit small enough to focus on a limited part of the text.

5.3 Embeddings and FAISS

The embedding model used is:

`sentence-transformers/all-MiniLM-L6-v2`

Each chunk is converted into a dense vector representation. These vectors are stored in a FAISS index. When a user asks a question, the question is embedded and compared with all chunk embeddings. The system retrieves the most similar chunks.

This semantic retrieval step is the core of the RAG pipeline. It allows the model to work with the most relevant passages instead of the entire corpus.

5.4 Language Model

The generative model used in the project is:

`meta-llama/Llama-3.1-8B-Instruct`

It is accessed through a Hugging Face endpoint and wrapped with `ChatHuggingFace`. This model was selected because it is instruction-tuned, meaning that it is designed to follow natural language instructions and produce structured answers.

This property is important because the model is instructed to answer only from the provided context and to avoid inventing information.

5.5 Prompt Design

The system prompt defines the model as a political journalist specialized in the French legislative elections of 1988. It asks the model to answer factually and rigorously from the context only.

The prompt also includes an explicit anti-hallucination instruction: if the answer is not present in the context, the model must say that it does not know. This is important because the project evaluates a closed-domain question-answering system, not a general chatbot.

6 Implementation

The project was implemented in Python on SSPCloud. The main libraries used are:

- `pandas` for data handling;
- `matplotlib` for exploratory plots;
- `scikit-learn` for TF-IDF and cosine similarity;
- `LangChain` for the RAG pipeline;

- `FAISS` for vector search;
- `Hugging Face` for embeddings and LLM access.

The implementation is organized into corpus loading, exploratory analysis, vector database creation, retrieval, generation, and evaluation.

The retrieval function takes a query and returns the top k most similar chunks from the FAISS database. The RAG function then builds a context from these chunks, sends it to the model, and returns the generated answer.

7 Evaluation

7.1 Comparison with a LLM-only Baseline

To evaluate the contribution of retrieval, we compare two settings:

- **RAG**: the model answers using retrieved chunks from the 1988 corpus;
- **LLM-only**: the model answers without retrieved context.

The evaluation uses five broad political questions based on documents relating to the French political manifestos of 1988:

1. What do the 1988 electoral programs say about unemployment in France?
2. How is education discussed in 1988?
3. What is said about France in 1988?
4. What is said about economic policy in France in 1988?
5. How is immigration discussed in France in 1988?

These questions were formulated to remain close to the corpus and to make the comparison between RAG and the LLM-only baseline more meaningful.

7.2 Groundedness Metric

We use a simple automatic groundedness metric. The idea is to measure how much of the generated answer is lexically supported by the retrieved context.

For an answer A and a context C, the score is:

$$\text{Groundedness}(A, C) = \frac{|W(A) \cap W(C)|}{|W(A)|},$$

where $W(A)$ is the set of cleaned words in the answer and $W(C)$ is the set of cleaned words in the retrieved context.

In words, the metric computes the proportion of words in the generated answer that also appear in the retrieved context. If the answer contains no valid word after cleaning, the score is set to zero to avoid division by zero.

This metric does not prove that the answer is fully correct. It does not capture all paraphrases and does not replace human evaluation. However, it is useful here because the goal of RAG is precisely to generate answers grounded in retrieved documents.

7.3 Evaluation Results

The RAG system obtains a higher groundedness score than the LLM-only baseline for all five questions.

Question	RAG score	LLM-only score
Unemployment	0.531	0.172
Education	0.521	0.167
France	0.594	0.103
Economic policy	0.690	0.154
Immigration	0.591	0.146
Average	**0.585**	**0.148**

Table 3: Groundedness comparison between RAG and LLM-only answers

On average, the RAG system reaches a groundedness score of 0.585, compared with 0.148 for the LLM-only baseline. This means that the RAG answers are much more lexically connected to the retrieved manifesto passages.

The result is expected but important: retrieval changes the behavior of the model. Without context, the LLM tends to answer from general knowledge or generic political reasoning. With RAG, the answer is more directly linked to the text of the corpus.

8 Limitations

First, the evaluation metric is simple. The groundedness score measures lexical overlap between the answer and the retrieved context. It is useful, but it does not fully measure factual correctness or semantic faithfulness. A response could reuse words from the context while still misinterpreting them. A stronger evaluation would require manually annotated reference answers.

Second, the project focuses only on 1988. This restriction is justified by computational constraints, but it limits the scope of the conclusions. Extending the vector database to 1973, 1978, 1981, and 1993 would allow comparison across elections.

Third, OCR noise may affect retrieval quality. Since the texts come from digitized documents, some transcriptions contain errors. These errors can influence both embeddings and generated answers.

Fourth, the results depend on technical choices such as chunk size, chunk overlap, embedding model, number of retrieved chunks, and prompt formulation. Different parameters could lead to different answers.

Despite these limits, the project shows that RAG is a relevant and practical NLP approach for exploring a historical political corpus.

9 Conclusion

This project implemented a Retrieval-Augmented Generation system for question answering over the 1988 French political manifestos from the Archelec corpus.

The pipeline includes corpus loading, exploratory data analysis, text cleaning, chunking, embeddings, FAISS semantic search, and generation with `Llama-3.1-8B-Instruct`. The corpus contains 3,628 documents and is split into 16,815 chunks.

The evaluation compares RAG with a LLM-only baseline on five political questions. The RAG system obtains a higher average groundedness score, 0.585 compared with 0.148 for the LLM-only baseline. This supports the main conclusion of the project: retrieval improves the connection between generated answers and the actual corpus.

Future work could extend the analysis to all available years, improve OCR cleaning, compare several embedding models, test different chunking strategies, and develop a human-annotated evaluation set for question answering.

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